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ROLLNO.: 20

COURSE: AI&DS

CLASS: BE

SUB: Computer Laboratory-III

Title: Simulation of Client Requests and Load Distribution Using Load Balancing Algorithms.

Code:

```
import random
from collections import defaultdict

class Server:
    def __init__(self, name):
        self.name = name
        self.connections = 0
    def handle_request(self):
        self.connections += 1
    def release_request(self):
        if self.connections > 0:
            self.connections -= 1

def round_robin(servers, requests):
    distribution = defaultdict(int)
    index = 0
    for _ in range(requests):
        server = servers[index]
        server.handle_request()
        distribution[server.name] += 1
        index = (index + 1) % len(servers)
    return distribution
```

```
def least_connections(servers, requests):
    distribution = defaultdict(int)
    for _ in range(requests):
        server = min(servers, key=lambda s: s.connections)
        server.handle_request()
        distribution[server.name] += 1
    return distribution

def random_selection(servers, requests):
    distribution = defaultdict(int)
    for _ in range(requests):
        server = random.choice(servers)
        server.handle_request()
        distribution[server.name] += 1
    return distribution

def reset_servers(servers):
    for server in servers:
        server.connections = 0

servers = [Server("Server-1"), Server("Server-2"), Server("Server-3")]
total_requests = 15

reset_servers(servers)
rr_result = round_robin(servers, total_requests)
print("Round Robin Distribution:")
for k, v in rr_result.items():
    print(k, ":", v)
```

```
Round Robin Distribution:
Server-1 : 5
Server-2 : 5
```

Server-3 : 5

```
reset_servers(servers)
lc_result = least_connections(servers, total_requests)
print("\nLeast Connections Distribution:")
for k, v in lc_result.items():
    print(k, ":", v)
```

Least Connections Distribution:

Server-1 : 5

Server-2 : 5

Server-3 : 5

```
reset_servers(servers)
random_result = random_selection(servers, total_requests)
print("\nRandom Selection Distribution:")
for k, v in random_result.items():
    print(k, ":", v)
```

Random Selection Distribution:

Server-2 : 7

Server-1 : 5

Server-3 : 3